Ruoshui Xu

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Website – Google Scholar – LinkedIn

RESEARCH INTERESTS

Bayesian Biostatistics; Adaptive Clinical Trial Design; Dose-Finding Methods; Bayesian Semiparametric Modeling; MCMC and Computational Statistics; Longitudinal and Clustered Health Data; Applied Statistics; Interdisciplinary Statistical Consulting; Experimental Design; Data Visualization

EDUCATION

Colorado State University, Fort Collins, CO — 05/2028(Expected)
Doctor of Philosophy, Statistics
Advisor: Dongzhou Huang, Ph.D.

Purdue University, West Lafayette, IN — 05/2021
Master of Science, Statistics

Purdue University, West Lafayette, IN — 05/2019
Master of Science, Environmental and Ecological Engineering

Beijing University of Civil Engineering and Architecture — 06/2016
Bachelor of Science, Environmental Science

PROFESSIONAL EXPERIENCE

Graduate Statistical Consultant — 01/2026 – 05/2026
Franklin A. Graybill Statistics and Data Science Laboratory — Fort Collins, CO

- Staffed four hours of weekly drop-in consultation sessions, advising graduate students and faculty across disciplines on research questions, data preparation, visualization, and common statistical analyses.
- Documented consultation outcomes, researched unresolved statistical questions, provided follow-up recommendations, and referred complex or longer-term projects to faculty statistical consultants when appropriate.

Data Analyst — 08/2021 – 07/2023
BioStone Animal Health LLC — Dallas, TX

- Collected, consolidated, and analyzed laboratory batch-testing data in Excel to evaluate diagnostic kit performance and prepare quality control reports.
- Redesigned and standardized the company's quality control reporting template, establishing a consistent format that was adopted for subsequent product batches.
- Led the organization and tracking of production and order data across diagnostic kit assembly, fulfillment, and distribution, and developed standard operating procedures to improve workflow consistency.
- Collaborated with laboratory, production, business, and international teams to support product development, operational coordination, and data-driven decision-making.
- Contributed to the development of an ELISA assay data analysis application by defining functional requirements, supporting feature design, validating analytical outputs, and incorporating user feedback.

Graduate Statistical Consultant — 08/2020 – 05/2021
Statistical Consulting Service, Purdue University — West Lafayette, IN

- Managed statistical consulting engagements for 20+ clients from diverse academic disciplines, conducting initial and follow-up meetings throughout the development of their research projects.
- Collaborated with clients, faculty advisors, and SCS faculty to clarify research objectives, evaluate study designs, and recommend appropriate data-analysis and interpretation strategies.
- Authored 100+ structured consultation reports documenting project backgrounds, statistical recommendations, progress, responsibilities, and next steps.

SELECTED RESEARCH PROJECTS

Bayesian Shape-Constrained Modeling of Dose–Utility Curves
R, MCMC, B-Splines, Bayesian Computation, Shape-Constrained Regression
- Developed a Bayesian logistic spline model for binary dose utility, using cubic B-spline edge bases and polyhedral-cone parameterization to enforce concavity on the logit scale.
- Derived and implemented a custom auxiliary-variable Gibbs sampler with truncated normal and gamma updates for intercept, linear-trend, and nonnegative spline coefficients.
- Evaluated posterior curve recovery against a known simulated utility function and a shape-constrained maximum-likelihood estimator using MCMC diagnostics and posterior uncertainty summaries.

Simulation-Based Comparison of CRM and BOIN Dose-Finding Designs
R, C++, Rcpp, Bayesian Statistics, Monte Carlo Simulation
- Implemented the Continual Reassessment Method (CRM) and Bayesian Optimal Interval (BOIN) design in C++ via Rcpp, incorporating grid-based Bayesian posterior updating, adjacent-dose transition constraints, and PAVA-based isotonic regression for final MTD selection.
- Designed and executed 6,000 simulated Phase I trials across three six-dose toxicity scenarios, with 30 patients per trial, to evaluate MTD selection, patient allocation, overdose exposure, toxicity risk, and computational efficiency.
- Developed an R analysis and visualization pipeline to aggregate operating characteristics and characterize the accuracy–safety trade-offs between model-based and model-assisted dose-finding strategies.

Bayesian Modeling of Hospital Fall Rates
R, rstanarm, Stan, MCMC, ggplot2
- Developed a Bayesian Poisson regression analysis of monthly hospital fall counts from a 26-month pre/post intervention study, incorporating intervention status, temporal effects, patient-day exposure, and patient-mix covariates.
- Fit and compared full, reduced, and intervention-focused model specifications using MCMC, examining chain behavior, posterior distributions, and posterior predictive simulations.
- Built a reproducible R workflow for missing-data assessment, exploratory visualization, Bayesian model fitting, diagnostic evaluation, and interpretation of intervention and demographic associations.

HONORS & AWARDS

Thomas J. And Eileen C. Boardman Statistical Consulting Award, CSU	2026
Graybill Award For Excellence In Linear Models, CSU	2025
Outstanding Thesis Award, BUCEA	2016
Merit-based Scholarship, BUCEA	2015, 2014, 2013

TEACHING EXPERIENCE

Colorado State University, Department of Statistics
Instructor of Record

Introduction to Applied Statistical Methods (STAT 301)	Spring 2026
Introduction to Applied Statistical Methods (STAT 301)	Fall 2025
Introduction to Applied Statistical Methods (STAT 301)	Summer 2025

Recitation Instructor

Statistics With Business Applications (STAT 204)	Summer 2024
Statistics With Business Applications (STAT 204)	Spring 2024, 2025
Statistics With Business Applications (STAT 204)	Fall 2023, 2024
General Statistics (STAT 201)	Spring 2024

Teaching Assistant

Statistical Consulting (STAA 556)	Summer 2026
Statistical Research - Design, Data, Methods (STAT 472)	Spring 2026
Design and Data Analysis for Researchers I (STAR 511)	Fall 2024

Purdue University, Department of Statistics
Teaching Assistant

Applied Regression Analysis (STAT 512)	Summer 2020
Applied Regression Analysis (STAT 512)	Spring 2020
Applied Regression Analysis (STAT 512)	Fall 2019

PUBLICATIONS

- Chen, B., Ma, W., Xu, R., & Zhang, J. (2019). Building an indicator-based assessment framework for adaptive capacity of drainage systems in Beijing. *Journal of Water and Climate Change*, **10**(2), 249–262. https://doi.org/10.2166/wcc.2018.156

METHODS & SOFTWARE

Research methods: Monte Carlo simulation studies; study and experimental design; clinical trial design; longitudinal healthcare data analysis; interdisciplinary statistical consulting; data collection, management, and quality control; reproducible computational workflows; cross-functional research collaboration

Statistical methods: Descriptive and inferential statistics; hypothesis testing; linear regression; analysis of variance; Bayesian generalized linear and hierarchical models; adaptive dose-finding methods; Markov chain Monte Carlo; shape-constrained regression; model comparison; data visualization

Software and platforms: R; C++; Rcpp; Stan; Microsoft Excel; LaTeX; Quarto; Git; WordPress; HTML

Languages: Mandarin Chinese (native); English (professional working proficiency)